

Database Technologies – Assignment 10

PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
kd1_99523_sandhya1> delimiter $$
kd1_99523_sandhya1> create procedure sp_rect(l int,w int)
-> begin
-> declare v_per INT;
-> declare v_area INT;
-> set v_per=2*(l+w);
-> set v_area=l*w;
-> select v_per as perimeter;
-> select v_area as area;
-> end;
-> $$
Query OK, 0 rows affected (0.01 sec)

kd1_99523_sandhya1> delimiter ;
kd1_99523_sandhya1> call sp_react(2,3);
ERROR 1305 (42000): PROCEDURE classwork.sp_react does not exist
kd1_99523_sandhya1> call sp_rect(2,3);
+-----+
| perimeter |
+-----+
|          10 |
+-----+
1 row in set (0.00 sec)

+-----+
| area |
+-----+
|      6 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
kd1_99523_sandhya1> DELETE FROM temp;
Query OK, 2 rows affected (0.02 sec)

kd1_99523_sandhya1> drop procedure calculate;
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhya1> delimiter $$
kd1_99523_sandhya1>
kd1_99523_sandhya1> create procedure calculate(IN num INT)
-> begin
-> declare square_value INT;
-> declare cube_value INT;
-> set square_value = num * num;
-> set cube_value = num * num * num;
-> insert into temp values(num, square_value, cube_value);
-> end $$
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhya1>
kd1_99523_sandhya1> delimiter ;
kd1_99523_sandhya1> call calculate(4);
Query OK, 1 row affected (0.02 sec)

kd1_99523_sandhya1> select * from temp;
+-----+-----+-----+
| num | square_value | cube_value |
+-----+-----+-----+
| 4 | 16 | 64 |
+-----+-----+-----+
1 row in set (0.00 sec)

kd1_99523_sandhya1> |
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are
 $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
kd1_99523_sandhya1> DELIMITER $$
kd1_99523_sandhya1>
kd1_99523_sandhya1> CREATE PROCEDURE convert_temperature(IN f DOUBLE, IN c D
DOUBLE)
-> BEGIN
-> DECLARE celsius DOUBLE;
-> DECLARE fahrenheit DOUBLE;
->
-> SET celsius = (f - 32) * 5 / 9;
-> SET fahrenheit = 9 / 5 * c + 32;
->
-> INSERT INTO temp_conversion VALUES(f, celsius, 'Fahrenheit to Cel
sius');
-> INSERT INTO temp_conversion VALUES(c, fahrenheit, 'Celsius to Fah
renheit');
-> END $$
Query OK, 0 rows affected (0.00 sec)

kd1_99523_sandhya1>
kd1_99523_sandhya1> DELIMITER ;
kd1_99523_sandhya1> CALL convert_temperature(100, 1);
Query OK, 1 row affected (0.02 sec)

kd1_99523_sandhya1> select * from temp_conversion;
+-----+-----+-----+
| input_temp | converted_temp | conversion_type |
+-----+-----+-----+
|          100 | 37.77777777777778 | Fahrenheit to Celsius |
|           1 | 33.8 | Celsius to Fahrenheit |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches

```
kd1_99523_sandhya1> CREATE TABLE inch_conversion
-> (
-> inches INT,
-> yards INT,
-> feet INT,
-> remaining_inches INT
-> );
Query OK, 0 rows affected (0.03 sec)

kd1_99523_sandhya1> DELIMITER $$
kd1_99523_sandhya1>
kd1_99523_sandhya1> CREATE PROCEDURE convert_inches(IN total_inches INT)
-> BEGIN
-> DECLARE yards INT;
-> DECLARE feet INT;
-> DECLARE inches INT;
->
-> SET yards = total_inches / 36;
-> SET feet = (total_inches % 36) / 12;
-> SET inches = total_inches % 12;
->
-> INSERT INTO inch_conversion
-> VALUES(total_inches, yards, feet, inches);
-> END $$
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhya1>
kd1_99523_sandhya1> DELIMITER ;
kd1_99523_sandhya1> CALL convert_inches(124);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> SELECT * FROM inch_conversion;
+-----+-----+-----+-----+
| inches | yards | feet | remaining_inches |
+-----+-----+-----+-----+
|      124 |      3 |      1 |           4 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
kd1_99523_sandhya1> CREATE TABLE product_result
-> (
->     num1 DOUBLE,
->     num2 DOUBLE,
->     product DOUBLE,
->     result VARCHAR(30)
-> );
Query OK, 0 rows affected (0.04 sec)

kd1_99523_sandhya1> DELIMITER $$
kd1_99523_sandhya1>
kd1_99523_sandhya1> CREATE PROCEDURE check_product(IN num1 DOUBLE, IN num2 DOUBLE)
-> BEGIN
->     DECLARE product DOUBLE;
->     DECLARE result VARCHAR(30);
->
->     SET product = num1 * num2;
->
->     SET result = CASE
->         WHEN product >= 100 THEN 'Equal to or greater than 100'
->         ELSE 'Less than 100'
->     END;
->
->     INSERT INTO product_result
->     VALUES(num1, num2, product, result);
-> END $$
Query OK, 0 rows affected (0.02 sec)

kd1_99523_sandhya1>
kd1_99523_sandhya1> DELIMITER ;
kd1_99523_sandhya1> CALL check_product(10, 10);
Query OK, 1 row affected (0.01 sec)

kd1_99523_sandhya1> select * from product_result;
+-----+-----+-----+-----+
| num1 | num2 | product | result |
+-----+-----+-----+-----+
| 10 | 10 | 100 | Equal to or greater than 100 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```